Project Phase 1: Data Requirements — Tower Running

Team 63: Carromboard

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1. Mini World Development

Our database, the **Towerrunning World Database (TWDB)**, captures the competitive world run by the Towerrunning World Association (TWA): stair races up skyscrapers, towers and outdoor staircases, the athletes who climb them, and the monthly world ranking built from their results.

The TWA calls itself the only worldwide governing body for stair racing. It regulates races, backs the Towerrunning Tour of the most prestigious races, manages the world ranking, and stages World and Continental Championships.

1.1 Domain Study & Idea Formulation

The table maps each part of the site to what it shows and what TWDB stores from it.

Site area	What the site shows	What TWDB captures
Races	More than 200 races a year, listed by month back to January 2014. Each entry shows date, race name, building, city, 3-letter country code, stairs, floors, male and female winner, and factor ("?" until assigned).	RACE, VENUE, NATION; winners derived from FINISHED

Site area	What the site shows	What TWDB captures
Race pages: R3313 PingAn, R3215 Metropolitan Sky Run, R3253 Empire State Building Run Up, R3303 Torino, R3395 Tampa, R3399 Affin Tower	Details block: race code, edition number, factor, category, male and female winner, contact person and e-mail, race link, result link, last edition's result link, video or picture links, free-text Additional Info ("Italian Championships", "no prerun due to rain"), previous edition and last factor. Venue block: name, country, city, stairs, floors, height difference, special info ("8 flat sections of about 60 m"), building photo and course records (Paul Crake (AUS), 9:33 in 2003). Category is T on Tour races (PingAn 200, Empire State 80, Torino 80) and R on regular races (Tampa, Affin Tower), whose last factor is 40.	RACE, COURSE, VENUE, PRECEDED_BY
Information for Event Organizers	Definition of a towerrun: stairs take more than 50% of the winner's time, at least 100 upward stairs, upward stairs ≥ downward stairs. Race modes: individual time trial, mass start, multi climb, tournament. Each race gets a factor from 40 to 280, fixed before the race. Factor 120/160/200 races are picked by the TWA at season start and need at least one previous edition; a new event can start as a Trial 60. Results (full name, gender, nationality and time of at least the top 30 men and top 30 women) are due within 7 days.	COURSE constraints, Factor rules, FINISHED completeness rule

Site area	What the site shows	What TWDB captures
Ranking, Ranking Basics and the July 2026 ranking file	Only registered athletes with a valid Athlete-ID are ranked. A score is the sum of the 8 best results in the last 12 months, published monthly as Excel files, with corrected re-issues ("October 2025 (corr)"). The July 2026 file has three sheets: 377 men, 194 women and 61 nations; the May 2025 file has the same layout and rules (294 men, 149 women, 53 nations), as does the Final 2025 file (338 men, 174 women, 54 nations). Each row shows rank, change in rank, name, nation, last month's score, score, change in score, races scored (regular, Tour, total), the best 8 results and the points won at every race in the window. On equal points the athlete with fewer races ranks higher; athletes still level share a rank. Points can be halves (591.5), so tied places share points. Races are numbered within each season: Tour races T01, T02, … and regular races R01, R02, … (up to T36 and R170 in season 2025); regular races are worth up to 40 points. A nation's result in a race is the points of its two best women and two best men, and its score is its 8 best race results.	RANKING_EDITION, RANKED_IN, NATION_RANKED_IN, SCORING_RULE, SEASON, PART_OF, derived Points and Current_Score

Site area	What the site shows	What TWDB captures
Towerrunning Tour 2026 and Tour 2025	Tour 2026 has 43 races from 30 Nov 2025 (Kuala Lumpur) to the 28 Nov 2026 Final in Shanghai, factors 60–240; ten are tagged as national championships, all at factor 80. Tour 2025 ran from 1 Dec 2024 to the 22 Nov 2025 Final. The ranking file starts both "Tour 2026" and "Season 2026" (regular races) right after the 2025 Final.	SEASON, PART_OF, FINAL_OF, CHAMPIONSHIP (national level)
Tour Final 2026 announcement	Shanghai Tower, 7th staging, factor 240, 119 floors, 3398 stairs, 552 vertical metres. 24 invitations: top 10 men and top 10 women of the October ranking plus 2 wild cards per gender; a declined invitation passes to the next-ranked athlete. Invitees get €600 travel and 2 nights' lodging. Prize money for places 1–6: €2300, €1800, €1300, €800, €500, €300. Applications close 17 Oct 2026. The November ranking is the year-end ranking, and the 2027 Tour starts in December 2026.	APPLIES_FOR, Is_Season_Final, season boundaries
Members / Registration	Yearly fee €24, or €18 for athletes aged 50+ or 20-, checked automatically from year of birth. Payment by card (MasterCard, VISA) or bank transfer; the Athlete-ID is mailed after payment, later for bank transfers. Login accepts username, e-mail or Athlete-ID.	REGISTERED_ATHLETE (three unique identifiers), MEMBERSHIP (weak), derived fee

Site area	What the site shows	What TWDB captures
Our Team	Presidium: President Daniel Cecetka, Sports Director Michael Reichetzeder. Representatives of 18 national associations (e.g. Towerrunning Slovakia, Towerrunning Germany, Malaysia Towerrunning Association), each with Facebook, e-mail and website links. Several representatives also race and win (Wai Ching Soh won the 2026 Metropolitan Sky Run and leads the July 2026 men's ranking).	PERSON with overlapping subclasses OFFICIAL and REGISTERED_ATHLETE, NATIONAL_ASSOCIATION
Statistics	Season statistics from 2009 onward and championship results: World Championships 2015 Doha, 2018 Taipei, 2024 Taipei; European 2012 Frankfurt, 2014 Vienna/Brno/Bratislava, 2016 and 2022 Warsaw; Asian/Oceanian 2015 Guangzhou, 2017 Taipei; PanAmerican 2016 Bogota. No 2020 season ranking because of the pandemic.	CHAMPIONSHIP (world, continental, national), DECIDED_AT (one championship can span several races)

Scope

In scope. Nations; people who appear in results or run the sport; registered athletes and their yearly memberships; national associations and TWA officials; venues; races with their courses, factors, seasons and editions; championships; the result lines of the TWA Results Form; the scoring matrix; monthly athlete and nation ranking editions; and athletes' applications and invitations for Tour races.

Out of scope, and why.

- News articles, the photo gallery and the Facebook feed: editorial content, not structured sports data. TWDB keeps only links (photo URLs, announcement URLs).

- Full result lists on organisers' websites: each race publishes them in its own format (Sportmaniacs, ChronoTrack, RunSignup, Checkpointspot). TWDB stores only what the standard TWA Results Form requires (name, gender, nationality and time) and keeps the organiser's page as a result link.
- Age-group categories: they appear only in occasional organiser notes (a new 90+ course record at the Empire State Building Run Up), and the 50+ and 20- columns of the May 2025 and July 2026 ranking files carry no age data. TWA uses those two bands only for the membership fee, which Fee_EUR already covers.
- Split times, chip timing and bib logistics: never published by TWA.
- Payment processing and card data: handled by the shop and gateway. TWDB stores payment method and status only, which is also safer.
- Public race-entry sign-ups: handled on each organiser's own website.
- Newsletter subscriptions and advertising banners: marketing, unrelated to competition data.

Assumptions not stated on the site: one championship per race, one recognised association per nation, no two courses of one race with the same number of upward stairs, and finishers sharing a place splitting its points equally (the ranking file's half points suggest this). The meaning of category T/R is inferred from race pages and the ranking file's race codes; the site never defines it.

1.2 User Identification & Interaction

Six groups use TWDB, each with a different level of access. The roles mirror who does what on the site today.

User	Who on the site	Reads	Writes
TWA Sports Director / ranking administrator	Michael Reichetzeder, who publishes rankings and announcements	Everything	Races, factors, seasons, scoring matrix, imported results, monthly ranking editions and corrections, Tour Final invitations
TWA Presidium	President Daniel Cecetka and fellow presidium members	Everything, mainly analysis reports	Championship records, wild-card invitations, approval of factor 120/160/200 races

User	Who on the site	Reads	Writes
National association representative	The 18 representatives on the Team page	Own nation's athletes, races and results	Proposed national races, national championship designations, own association profile
Event organiser / timing company	Organisers such as PT Cakrawala Jakarta Kreatif (Jakarta Vertical Run)	Own races, applications for elite packages	Race and venue details before the race; the Results Form within 7 days after it
Registered athlete	Anyone holding an Athlete-ID	Own profile, results, score and rank; full calendar one year ahead	Registration and yearly membership, profile edits, applications for Tour races, accept/decline invitations
Public visitor	Anyone browsing the site	Races in the next few weeks, published rankings, statistics, championship results	Nothing

The access split follows the site itself: the Races page states that the complete calendar a year ahead is visible only in a registered athlete's profile, while the public sees the coming weeks (Races).

1.3 Purpose of the Database

TWDB replaces a spread of WordPress posts, PDFs and monthly Excel files with one consistent store from which the ranking is computed rather than hand-assembled. Today results arrive on an Excel Results Form, rankings are published as separate monthly .xlsx files, and the Ranking page itself warns that the automatically generated dashboard ranking is still a work in progress (Ranking).

The site already shows the problems that free-text, per-page data causes:

- **Same person, different spellings.** The Austrian representative appears as "Klaus Hausleitner" on the Team page and in the ranking, but as winner "Klaus Hsusleitner" on the Races page; the ranking also spells Görge Heimann as "Goerge Heimann". A file

system cannot tell these are one athlete; TWDB links every result to one PERSON key.

- **Implausible values.** Affin Tower in Kuala Lumpur is listed with 1004 stairs and "4242 Floors". A domain check (floors ⩽ stairs, integrity rule 2) rejects this at entry.
- **Labels that go stale.** Metropolitan Sky Run is a factor-120 Tour race in 2026, yet its page still shows category R, carried over from its regular 2025 edition. TWDB derives the category from the factor, so the two cannot disagree.
- **Contradictory facts.** The About text says nine national associations; the Team page lists 18. Stored once, a count is always derived, never retyped.
- **Manual corrections.** Rankings such as "October 2025 (corr)" were re-issued by hand. In TWDB a corrected time cascades automatically to places, points and scores, and a new ranking edition is published.

What a DBMS offers that the current pages and spreadsheets do not:

1. **Integrity enforced at entry**: only valid Athlete-IDs earn points, Tour factors must be set before the race, courses must meet the 100-stair and 50% rules.
2. **Derived data computed, not typed**: points from the scoring matrix, the 8-best rolling score, age-based fees, nation scores.
3. **History kept**: every monthly ranking edition stays queryable, so an athlete's rank in October 2025 is a query, not a hunt through downloads.
4. **Concurrent, controlled access**: organisers upload results, athletes apply for races and the Sports Director publishes rankings at the same time, each limited to what their role may change.
5. **Cross-entity questions answered in one query**: which officials also win races, which nations top the nations ranking, which venues are the fastest per vertical metre.

2. Database Requirements

TWDB has 13 entity types (2 weak, 2 subclasses) and 15 relationship types: 14 binary (one of them recursive) and 1 ternary. The checklist below maps each assignment requirement to where it is met.

Requirement	Met by
⩾ 5 entity types	13 entity types
Entity with two key attributes	NATION (Nation_Code, Nation_Name); REGISTERED_ATHLETE (Athlete_ID, Username, Email)
2 weak entity types	COURSE (owner RACE), MEMBERSHIP (owner REGISTERED_ATHLETE)

Requirement	Met by
⩾ 8 relationship types	15 relationship types
One n ⩾ 3 relationship	FINISHED (PERSON, COURSE, NATION)
⩾ 1 subclass	REGISTERED_ATHLETE and OFFICIAL under PERSON
⩾ 2 composite attributes	Name, Address, Contact, Payment
⩾ 2 multivalued attributes	Contact_Links, Photo_URLs, Media_Links
⩾ 2 derived attributes	Age, Category, Current_Score, Fee_EUR, Gender_Place, Points, Is_Evaluated, Male/Female_Winner, Rank_Change
Bonus: candidate, super and alternate keys	REGISTERED_ATHLETE
Bonus: ternary as binaries + weak entity	FINISHED
Bonus: same entity in two roles	PRECEDED_BY (RACE as previous and next edition)

Notation: **key** in bold, *partial key* in italics, {…} = composite, [multi] = multivalued, (derived) = derived.

2.1 Strong entity types

Entity	Key(s)	Other attributes	Grounding
NATION	**Nation_Code**, **Nation_Name** (both unique)	Continent	Race pages use 3-letter codes (CHN, SVK, MAS, NED); the Nations ranking lists name and code together (China, CHN)
PERSON (superclass)	**Person_ID**	Name {First_Name, Last_Name}; Gender; Date_of_Birth; Age (derived)	Results Form needs full name, gender, nationality; unregistered finishers still count for the nations ranking

Entity	Key(s)	Other attributes	Grounding
NATIONAL_ASSOCIATION	**Assoc_ID**	Assoc_Name; Founded_Year; Contact_Links [multi] (website, Facebook, e-mail)	Team page shows each association with several links
VENUE	**Venue_ID**	Venue_Name; Venue_Type; Address {Street, City, Postal_Code}; Building_Height_m; Photo_URLs [multi]; Course_Record_Male, Course_Record_Female (derived)	Race page Venue block: name, country, city, building photo, course records; organisers send 2–4 pictures
RACE	**Race_Code**	Race_Name; Race_Date; Duration_Days; Edition_No; Factor; Category (derived); Status; Evaluation_Status; Contact {Contact_Person, Contact_Email}; Additional_Info (free text); Media_Links [multi] (race link, result link, TWA announcement, video, pictures); Results_Received_On; Male_Winner, Female_Winner (derived)	Race pages R3313, R3215, R3253; the ranking file marks races "not ev." (not evaluated) and "corr." (corrected)
SEASON	**Season_Year**	Start_Date; End_Date	Tour 2025 ran 1 Dec 2024 – 22 Nov 2025; the ranking file starts Tour 2026 and Season 2026 right after it; the 2027 season starts in December 2026. The Final 2025 ranking ends exactly at T36 (the Final) and R170, so a season closes with its Final
CHAMPIONSHIP	**Championship_ID**	Title; Level; Region; Year	Statistics page (World, European, Asian/Oceanian, PanAmerican) and national tags on the Tour page and in Additional Info

Entity	Key(s)	Other attributes	Grounding
RANKING_EDITION	**Ranking_ID**	Ranking_Month; Ranking_Year; Revision_No; Is_Season_Final; Published_On	Monthly files with Males, Females and Nations sheets, incl. "Final 2025" and " (corr)" re-issues
SCORING_RULE	**{Factor, Place}**	Points	TWA Scoring Matrix: points per factor column and place

2.2 Subclasses and weak entity types

Specialisation of PERSON into REGISTERED_ATHLETE and OFFICIAL is **overlapping** and **partial**. Overlapping because officials also race: Wai Ching Soh (Malaysia representative) won the Metropolitan Sky Run and Jindrich Schick (Czech representative) won the MS-UK Tower Walk. Partial because many result-list finishers never register.

Entity	Kind	Key	Attributes
REGISTERED_ATHLETE	Subclass of PERSON	**Athlete_ID**, **Username**, **Email** (each unique); inherits Person_ID	Registered_On; Current_Score (derived)
OFFICIAL	Subclass of PERSON	inherits Person_ID	Official_Role; Since_Year
COURSE	Weak, owner RACE via HAS_COURSE	*Course_Name* → full key (Race_Code, Course_Name)	Race_Mode; Upward_Stairs; Downward_Stairs; Floors; Vertical_Metres; Flat_Section_m; Special_Info; Stair_Time_Share_Pct; Is_Evaluated (derived)
MEMBERSHIP	Weak, owner REGISTERED_ATHLETE via HOLDS	*Member_Year* → full key (Athlete_ID, Member_Year)	Payment {Payment_Method, Payment_Reference}; Paid_On; Status; Fee_EUR (derived)

The site prints "No stairs pos", "No floors pos", "Height df pos" and "Special Info" under Venue; TWDB stores them on COURSE as Upward_Stairs, Floors, Vertical_Metres and Special_Info (section 2.5 explains why). COURSE is weak because "Elite" or "Triple Climb" means nothing without its race; a race offering several distances (Triple Climb, Power Hour) has several courses, and only the longest is scored (Ranking Basics). MEMBERSHIP is weak because the fee is "basically valid for a calendar year" for one athlete (Members).

2.3 Relationship types

#	Relationship	Participants (cardinality)	Participation	Attributes / meaning
R1	CITIZEN_OF	PERSON N : 1 NATION	PERSON total; NATION partial	Current nationality
R2	LOCATED_IN	VENUE N : 1 NATION	VENUE total; NATION partial	Country of the building
R3	HELD_AT	RACE N : 1 VENUE	RACE total; VENUE partial	One venue per race; a venue hosts many editions
R4	HAS_COURSE (identifying)	RACE 1 : N COURSE	both total	Every race offers at least one course
R5	HOLDS (identifying)	REGISTERED_ATHLETE 1 : N MEMBERSHIP	both total	An Athlete-ID exists only after a paid membership
R6	PART_OF	RACE N : 1 SEASON	RACE total; SEASON partial	Season_Race_No: the race's number within its season and category (T05, R03)
R7	FINAL_OF	SEASON 1 : 1 RACE	both partial	The season's Tour Final; set once announced, since a season must exist before its races are added
R8	DECIDED_AT	CHAMPIONSHIP 1 : N RACE	CHAMPIONSHIP total; RACE partial	The race(s) whose results decide the championship: the Torino race is the Italian Championships; the 2014 European Championships spanned races in Vienna, Brno and Bratislava
R9	GOVERNS	NATIONAL_ASSOCIATION 1 : 1 NATION	ASSOCIATION total; NATION partial	One recognised association per country
R10	REPRESENTS	OFFICIAL N : 1 NATIONAL_ASSOCIATION	OFFICIAL partial (presidium members represent none); ASSOCIATION total	National representative
R11	RANKED_IN	REGISTERED_ATHLETE M : N RANKING_EDITION	ATHLETE partial; EDITION total	Rank_Position, Score_Points, Regular_Races, Tour_Races; Total_Races, Rank_Change, Score_Change (derived)
R12	NATION_RANKED_IN	NATION M : N RANKING_EDITION	NATION partial; EDITION total	Same attributes as RANKED_IN, for the nations ranking
R13	APPLIES_FOR	REGISTERED_ATHLETE M : N RACE	both partial	Applied_On, Application_Status, Travel_Reimbursement_EUR
R14	PRECEDED_BY (recursive)	RACE (next edition) 1 : 1 RACE (previous edition)	both partial	Links editions of the same event
R15	FINISHED (ternary)	PERSON, COURSE, NATION	all partial	Finish_Time, Result_Status, Gender_Place (derived), Points (derived)

FINISHED records one line of a results sheet: which person, on which course, racing for which nation. For a given (PERSON, COURSE) there is exactly one NATION, so the relationship's key is (Person_ID, Race_Code, Course_Name). The NATION here is the

nationality on that day's results sheet, which can differ from the current CITIZEN_OF nation, and it is what the nations ranking uses.

2.4 Domains and integrity constraints

Attribute domains

- Nation_Code: exactly 3 uppercase letters, using the codes the site uses (IOC-style: MAS, NED, SLO rather than ISO MYS, NLD, SVN).
- Gender ∈ {M, F}, matching the separate male and female rankings.
- Race_Code: "R" followed by 1–6 digits (e.g. R3313).
- Factor ∈ {40, 60, 80, 120, 160, 200, 240, 280}, or NULL while unassigned (shown as "?"). Ranking Basics lists 40–280 without 240, but the 2026 Tour Final is a 240 race, so 240 is included. 280 has been used too: the Taipei 101 race of May 2024, the year of the World Championships there.
- Category ∈ {T, R}: T = Towerrunning Tour race, R = regular ranking race.
- Season_Race_No: positive integer, shown with the category letter (T05, R03).
- Race_Mode ∈ {Individual time trial, Mass start, Multi climb, Tournament}, the modes the organisers' guide names.
- Status (RACE) ∈ {Announced, Confirmed, Postponed, Cancelled, Completed}; Evaluation_Status ∈ {Pending, Evaluated, Corrected, Not evaluated}.
- Duration_Days ∈ 1–3 (The Sky Race runs 2 days, Acapulco Grand Prix 3 days).
- Level ∈ {World, Continental, National}; Region ∈ {Europe, Asia/Oceania, Pan-America}, required only when Level = Continental.
- Payment_Method ∈ {Card, Bank transfer}; MEMBERSHIP Status ∈ {Pending, Active, Expired}.
- Application_Status ∈ {Applied, Invited, Wild card, Accepted, Declined, Not selected}.
- Result_Status ∈ {FIN, DNF, DSQ}; Finish_Time in seconds, > 0, required when FIN.
- Points and Score_Points ⩾ 0, in steps of 0.5, because tied places share points.
- Rank_Position ⩾ 1; Ranking_Month ∈ 1–12; Revision_No ⩾ 0 (0 = original, 1+ = corrected re-issue).
- Email: valid address format; Username 3–30 characters.

Derived attributes and their rules

- Age = whole years between Date_of_Birth and today.
- Fee_EUR = 18 if (Member_Year − birth year) ⩾ 50 or ⩽ 20, else 24, as the registration page applies it automatically from year of birth.
- Category = T if Factor ⩾ 60, otherwise R (regular races score at factor 40, or have no factor yet).

- Is_Evaluated = true for the race's course with the most Upward_Stairs (the longest offered distance), false for all others.
- Gender_Place = position of the person's Finish_Time among FIN results of the same gender on the same course; equal times share a place.
- Points = SCORING_RULE.Points for (race Factor, Gender_Place) if the course Is_Evaluated and the race's Evaluation_Status is Evaluated or Corrected; otherwise 0. Finishers sharing a place split the points of the places they cover equally. Every finisher gets Points, because the nations ranking counts unregistered athletes; only REGISTERED_ATHLETEs with an Active membership for that year have them counted in Current_Score and RANKED_IN.
- Current_Score = sum of the 8 highest ranking-eligible Points from races in the last 12 months.
- A nation's result in a race = Points of its two best women plus its two best men there (by race-day nation). Its score = sum of its 8 best race results in the last 12 months.
- Total_Races = Regular_Races + Tour_Races. Rank_Change and Score_Change = differences from the same athlete's or nation's row in the previous edition.
- Male_Winner / Female_Winner = the person with Gender_Place 1 on the evaluated course. Course_Record_Male / Female = the fastest evaluated FIN time of that gender at the venue.

Integrity rules, from the site's own regulations

1. A course must have Upward_Stairs ⩾ 100, Downward_Stairs ⩽ Upward_Stairs, and Stair_Time_Share_Pct > 50 (Organizers Info).
2. Floors ⩽ Upward_Stairs and Vertical_Metres > 0 (sanity checks that would catch "4242 floors").
3. No two courses of the same race have the same Upward_Stairs, so the evaluated (longest) course is always unique.
4. A Tour race's Factor (⩾ 60) must be set before Race_Date and cannot change on or after it. A regular race's Factor stays NULL until the race is evaluated, then becomes 40.
5. A race with Factor 120, 160 or 200 must have a PRECEDED_BY edition; a first edition can be at most a Trial 60.
6. A race can be marked Evaluated only if Results_Received_On ⩽ Race_Date + 7 days and its sheet lists at least the top 30 men and top 30 women (or all finishers if fewer).
7. Each person has at most one FINISHED entry per course.
8. Every race's Race_Date lies between its season's Start_Date and End_Date, and Season_Race_No is unique within a season and category.
9. At most one MEMBERSHIP per athlete per year; Status = Active requires Paid_On.

10. In each RANKING_EDITION, positions follow Score_Points (highest first), then Total_Races (fewest first); rows still level share a Rank_Position, so positions are not unique. In July 2026, 150 men share 228th place on 0 points.
11. Tour Final invitations: at most 24 per final — the top 10 men and top 10 women of that season's October ranking plus at most 2 wild cards per gender.
12. A previous edition's Race_Date is earlier than the next edition's.

Enforcement. Keys and uniqueness via PRIMARY KEY / UNIQUE, including UNIQUE (Race_Code, Upward_Stairs) on COURSE for rule 3; domains via CHECK and enumerated types; references via FOREIGN KEY with stated delete rules (CASCADE from RACE to COURSE and from athlete to MEMBERSHIP, RESTRICT from COURSE to FINISHED, SET NULL on PRECEDED_BY). Cross-table rules (5, 6, 8, 11) are enforced by triggers or by the application inside one transaction.

2.5 Redundancy check

- A race's country is not stored on RACE; it is read through VENUE → NATION, though the site prints it on every race card.
- Winners, places, points, scores, categories, ages and fees are derived, never typed in, so they cannot disagree with the times or the factor. Category shows why: the site's stored R on the 2026 Metropolitan Sky Run contradicts its factor of 120.
- Stairs, floors, height difference and special info live on COURSE, not VENUE, even though the site prints them under Venue. The "pos" fields count only the upward climb of that race's course, which differs from the building itself: Shanghai Tower is 632 m tall but its race climbs 552 m; Ping An Finance Centre is 599 m tall but its course climbs 547 m. Storing them on VENUE would force editing the building whenever a course changes.
- The site's "Last factor" and "Result Link Old" fields are not stored; they are the previous edition's Factor and result link via PRECEDED_BY. Course records are not stored either: they are the fastest evaluated times at the venue, with historical records (e.g. Paul Crake (AUS), 9:33 in 2003 at the Empire State Building) loaded as past results.
- The ranking file's "Last", "Dev." and "d+-" columns are not stored; they come from the previous edition's rows.
- Kept on purpose: Score_Points and Rank_Position on RANKED_IN and NATION_RANKED_IN look derivable but are frozen monthly snapshots. The 12-month window moves and results get corrected, so old rankings cannot be recomputed later.
- Kept on purpose: FINISHED's NATION versus CITIZEN_OF. One is nationality on race day, the other today's.

2.6 Bonus

Keys of REGISTERED_ATHLETE. The site's login accepts "Username, E-mail or Athlete ID", so each identifies one athlete.

- Candidate keys: {Athlete_ID}, {Username}, {Email}, and the inherited {Person_ID}. Each is unique and minimal.
- Primary key: Athlete_ID, the identifier printed on rankings.
- Alternate keys: Username, Email, Person_ID.
- Super keys: any set containing a candidate key, e.g. {Athlete_ID, Email} or {Username, Last_Name, Gender}. These are unique but not minimal.

FINISHED as binary relationships and a weak entity. Replace the ternary with a weak entity RESULT (Finish_Time, Result_Status, Gender_Place, Points) and three binary relationships:

1. ACHIEVED: PERSON 1 : N RESULT, identifying, RESULT total.
2. RECORDED_ON: COURSE 1 : N RESULT, identifying, RESULT total.
3. RACED_FOR: RESULT N : 1 NATION, RESULT total.

RESULT's key is (Person_ID, Race_Code, Course_Name), the same as the ternary's. Three plain binaries without RESULT (PERSON–COURSE, PERSON–NATION, COURSE–NATION) would lose information: PERSON–NATION cannot say for which race the nation applied, and joining the three can produce combinations that never happened.

Same entity in two roles. PRECEDED_BY relates RACE (next edition) to RACE (previous edition), as on the R3313 page's "Previous edition" and "Last factor" fields. Rule 5 depends on it.

3. Functional Requirements

3.1 Retrieval Operations

3.1.1 Selection

- S1. Retrieve the complete rows of all races in season 2026 with Factor ⩾ 120.
- S2. Retrieve the complete rows of all courses with more than 3000 upward stairs (e.g. Shanghai Tower, 3398; PingAn Finance Centre, 3201).

3.1.2 Projection

- P1. Display Athlete_ID, name and nation of every athlete with an Active membership for 2026.

- P2. For a venue chosen by name, display Race_Name, Race_Date and Factor of every race held there.

3.1.3 Aggregate

- A1. Maximum Vertical_Metres over all evaluated courses (the tallest race on the calendar).
- A2. Number of Tour races per factor in season 2026 (COUNT grouped by Factor).
- A3. Fastest (MIN) Finish_Time per gender across all editions of the Empire State Building Run Up, i.e. the course record.
- A4. Average Upward_Stairs of evaluated courses per country.

3.1.4 Search

- Q1. Find athletes whose first or last name contains a typed fragment, ignoring case: "chom" matches Chomanicova.
- Q2. Find races whose name contains "Tower Run": matches Varso Tower Run, DC Tower Run, Nivy Tower Run and others.

3.2 Analysis Report

Each report joins several entity types; none is answerable from one table.

1. **Nations ranking** for a chosen month. For each nation, compute its result at every evaluated race (points of its two best women and two best men, by race-day nation, registered or not), sum its 8 best race results in the past 12 months, and show rank, score and change since the previous edition. Joins FINISHED, PERSON, COURSE, RACE, SCORING_RULE, NATION, NATION_RANKED_IN, RANKING_EDITION.
2. **Tour Final invitation list.** From the season's October RANKING_EDITION, list the top 10 per gender who applied for that season's Final, with nation, membership status and Application_Status; flag anyone without an Active membership. Joins RANKED_IN, RANKING_EDITION, REGISTERED_ATHLETE, MEMBERSHIP, APPLIES_FOR, RACE, FINAL_OF, SEASON.
3. **Officials who race.** Every OFFICIAL who is also a REGISTERED_ATHLETE, with their association, its nation, number of race wins and best-ever rank. Joins OFFICIAL, REPRESENTS, NATIONAL_ASSOCIATION, GOVERNS, NATION, FINISHED, RANKED_IN.
4. **Venue speed table.** For each venue: number of editions, course record per gender, and record vertical speed in metres per minute (Vertical_Metres ÷ record time). Joins VENUE, RACE, COURSE, FINISHED, NATION.

5. **Results compliance.** Races in a season whose results arrived more than 7 days after the race, or list fewer than 30 finishers of a gender when more competed, with their Evaluation_Status and the organiser's contact e-mail. Joins RACE, COURSE, FINISHED, PART_OF, SEASON.

3.3 Modification Operations

(a) Insertions with integrity checks

- I1. **Register an athlete.** Insert PERSON, REGISTERED_ATHLETE and the first MEMBERSHIP in one transaction. Reject a duplicate Username or Email, a missing Date_of_Birth, or a nation code not in NATION. Fee_EUR is computed, not entered. Card payments start Active; bank transfers start Pending, and the Athlete-ID is not released or points-eligible until verified.
- I2. **Enter one line of a Results Form.** Reject if the race is not yet past, the course or nation does not exist, the person already has a result on that course, Finish_Time ⩽ 0 for a FIN, or the line's gender differs from the PERSON's. If a PERSON with the same full name, gender and nation already exists, the line is linked to that person instead of creating a duplicate; near-miss spellings ("Hsusleitner" vs "Hausleitner") are merged manually by the administrator.
- I3. **Add a race.** Race_Code must match R + digits and be new; the venue must exist; Factor must be NULL or in the domain; Factor 120/160/200 needs a previous edition; the race joins the season its date falls in and takes the next free Season_Race_No for its category; at least one valid course (⩾ 100 upward stairs) is inserted in the same transaction.

(b) Updates with cascading rules

- U1. **Correct a finish time.** Recompute Gender_Place for that gender on that course, then Points, then the Current_Score of every affected athlete and the score of every affected nation. If the race falls inside an already published ranking, publish a new RANKING_EDITION with Revision_No + 1 (the site's "(corr)" files) and leave the original untouched.
- U2. **An invitee declines the Tour Final.** Set their Application_Status to Declined. The next athlete of the same gender in the October ranking who applied and holds an Active membership becomes Invited with €600 travel reimbursement, keeping the total at ⩽ 24.
- U3. **Postpone a race** (the Tour page marks Macau's Oxfam Tower Run "date changed"). Update Race_Date and set Status to Postponed. If the new date falls in another season, move PART_OF to that season with a new Season_Race_No; re-check that the factor is still fixed before the new date.

- U4. **Verify a bank transfer.** Set MEMBERSHIP Status from Pending to Active and fill Paid_On. The Athlete-ID is released and that athlete's results from the same year become points-eligible, so Points and Current_Score are recomputed.

(c) Deletion with referential integrity

- D1. **Delete a cancelled race.** Refuse if any FINISHED row exists on its courses. Refuse if it is the only race of a championship (total participation). If it is a season's Tour Final, clear that season's FINAL_OF link first. Otherwise delete its COURSE rows (weak entities) and APPLIES_FOR rows, notify applicants, and re-link PRECEDED_BY so the next edition points to the edition before the deleted one, keeping integrity rule 5 valid.

Use of Generative AI

AI was used for researching the domain and generating the document draft. We then went through the whole document, cross-checking the references and the results and iterating over it if something seemed wrong.

Sources

- Towerrunning home page and About us
- Races calendar
- Race pages: R3313 International Vertical Marathon PingAn, R3215 Metropolitan Sky Run, R3253 Empire State Building Run Up, R3303 Vertical Grattacielo Regione Piemonte, R3395 T2T Climb Tampa, R3399 Oregeon Tower Run @ Menara Affin
- Towerrunning Tour 2026 and Towerrunning Tour 2025
- Ranking and the TWA Ranking July 2026 file (.xlsx) and TWA Ranking May 2025 file (.xlsx) and TWA Ranking Final 2025 file (.xlsx)
- Ranking Basics (PDF)
- Information for Event Organizers (PDF)
- Members / Registration
- Our Team
- Statistics
- News, including the Tour Final 2026 announcement